

Notes: Drechsler, Savov, and Schnabl (QJE, 2017)

The Deposits Channel of Monetary Policy

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1 Big picture

- **Question.** How does monetary policy affect banks through the **deposit side** of their balance sheets, and can deposit-market power create a separate transmission channel from the standard reserve-based bank-lending channel?
- **Setting.** The paper studies U.S. commercial banks using aggregate time-series evidence and micro evidence from **branches, banks, counties, and small-business lending data**. The central mechanism is tested in local deposit markets, defined mainly at the **county** level.
- **Core challenge.** When the Fed funds rate rises, deposits may fall either because banks deliberately contract deposit supply or because their lending opportunities weaken. The paper must distinguish a **direct deposit-supply effect** from confounding changes in loan demand or local credit conditions.
- **Main theoretical contribution.** The paper develops a model in which households demand liquidity from **cash and deposits**, while banks have **market power over deposits**. When the Fed funds rate rises, cash becomes more expensive to hold, deposit demand becomes less elastic, and banks optimally widen deposit spreads. This contracts deposits and, because deposits are special funding, also contracts lending.
- **Main empirical strategy.** The paper exploits **geographic variation in local deposit-market concentration**. It compares different branches of the **same bank** located in counties with different concentration, so bank-level lending opportunities are differenced out. For lending, it compares **different banks within the same county**.
- **Why it matters.** This mechanism provides a new foundation for the bank-lending channel **without relying on required reserves**. It also links monetary policy to the supply of **safe and liquid assets**, helping explain movements in the liquidity premium.
- **Main empirical question.** When the Fed funds rate rises, do banks in more concentrated deposit markets raise deposit spreads more, lose more deposits, and contract lending more than otherwise similar banks?
- **Main empirical results.**
 - In aggregate, banks raise deposit rates far less than one-for-one with the Fed funds rate, so the **deposit spread widens strongly**.
 - Within a bank, a 100 bps increase in the Fed funds rate causes branches in high-concentration markets to raise spreads by about **14 bps more on savings deposits** and **7 bps more on time deposits** than low-concentration branches.
 - The same 100 bps increase causes **66 bps larger deposit outflows** at high-concentration branches.
 - The implied semi-elasticity of deposits with respect to deposit spreads is about -5.3 .
 - For lending, a 100 bps increase in the Fed funds rate combined with a one-standard-deviation increase in bank-level concentration reduces new small-business lending by about **291 bps**.
 - Counties served by banks that raise deposits in concentrated markets experience **lower lending, lower employment growth, and lower wage-bill growth** after monetary tightening.

- The average bank-level deposit spread beta is **0.54**, and **0.61** for the largest 5% of banks.
- For large banks, a typical **400 bps hiking cycle** implies roughly a **1,458 bps reduction in deposits** and a **995 bps reduction in lending**, or about **\$1.35 trillion** and **\$763 billion**, respectively.
- The aggregate deposit spread and the T-bill liquidity premium move together very closely, with correlation around **90%**.
- **Economic takeaway.** Monetary policy changes the **market power of banks over deposits**. When rates rise, banks can widen deposit spreads, shrink the supply of deposits, and thereby tighten credit supply and liquidity supply in the broader economy.

2 Data and measurement (key objects)

2.1 Aggregate and micro data sources

- **FDIC branch data.** Annual branch-level deposits for the universe of U.S. bank branches, from **June 1994 to June 2014**. These data provide branch identifiers, locations, and deposit quantities.
- **RateWatch deposit-rate data.** Weekly branch-level deposit rates, from **January 1997 to December 2013**. Coverage is about **54%** of all U.S. branches as of 2013. The paper keeps branches that actively set rates.
- **Call Reports.** Quarterly bank balance sheet and income statement data for all U.S. commercial banks, from **1994 to 2013**.
- **NCRC small-business lending data.** Bank-county annual data on new small-business loans, from **1997 to 2013**.
- **Monetary-policy data.** Fed funds target rate, one-year T-bill rate, and Fed funds futures.
- **County data.** U.S. Census and County Business Patterns for demographics; Bureau of Labor Statistics for county employment and wage bill.

2.2 Markets, banks, branches, products, and concentration

- **Local market definition.** The relevant deposit market is a **county**. Market power is measured using the county-level deposit-market Herfindahl index.
- **Branch-level concentration.** Each branch is assigned the HHI of the county where it is located; this is **Branch-HHI**.
- **Bank-level concentration.** A bank's **Bank-HHI** is the deposit-weighted average of Branch-HHI across all its branches.
- **County exposure to the deposits channel.** A county's **County-HHI** is the lending-share-weighted average Bank-HHI of banks operating in that county.
- **Deposit products.** The empirical analysis focuses on the two most common branch-level products:

- **\$25K money market accounts** as the main savings-deposit product,
- **\$10K 12-month CDs** as the main time-deposit product.

2.3 Key descriptive objects

- **Core deposits.** Checking, savings, and small time deposits are the main stable funding source of banks. In 2014, they totaled about **\$9.3 trillion**, or roughly **79% of bank liabilities**.
- **Aggregate deposit pricing fact.** For every 100 bps increase in the Fed funds rate, the spread between the Fed funds rate and the average deposit rate rises by about **54 bps**.
- **Liquidity ranking of deposit types.** Savings and checking deposits are highly liquid; small time deposits are less liquid because they are locked in for term.
- **Aggregate quantity responses.** Year-over-year growth in core deposits is strongly negatively related to changes in the Fed funds rate. The negative relation is strongest for savings and checking deposits, while small time deposits can move the other way as households substitute into less liquid products.
- **Concentration dispersion.** Branch-HHI ranges from roughly **0.06** to **1**, so there is large cross-county variation in local market power.
- **County characteristics by concentration.** Low-HHI counties are larger, richer, and more educated on average; high-HHI counties are smaller, poorer, older, and less educated.
- **Branch and bank size.** The average branch in the FDIC sample holds about **\$67 million** in deposits; the average branch in the RateWatch sample is larger, around **\$143 million**. Banks in low- and high-concentration areas are similar enough that within-bank and within-county comparisons are informative.

2.4 Unobservables and timing

- **Key omitted variable concern.** In aggregate, changes in deposits may reflect changes in **lending opportunities** rather than direct shifts in deposit supply.
- **Timing of deposit-rate adjustment.** Deposit spreads are measured quarterly in the main branch regressions and weekly in the event study.
- **Timing of deposit quantities.** Branch-level deposit flows are measured annually.
- **Fund reallocation inside the bank.** The identifying assumption is that banks can raise deposits at one branch and lend them elsewhere, so branch-level deposit pricing can be analyzed while holding bank-level lending opportunities fixed.
- **Other sources of market power.** Besides concentration, market power can come from depositor inattentiveness, low financial sophistication, and low willingness to switch banks.

3 Models and empirical specifications (all equations + interpretation)

3.1 Household utility and liquidity services

The representative household values both final wealth and liquidity services:

$$u(W_0) = \left[W^{\frac{\rho-1}{\rho}} + \lambda l^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}}. \quad (1)$$

Liquidity is itself produced from cash and deposits:

$$l(M, D) = \left(M^{\frac{\epsilon-1}{\epsilon}} + \delta D^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}}. \quad (2)$$

Deposits are a CES aggregate across banks:

$$D = \left(\frac{1}{N} \sum_{i=1}^N D_i^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}. \quad (3)$$

Interpretation.

- ρ is the elasticity of substitution between wealth and liquidity; the paper focuses on $\rho < 1$, so wealth and liquidity are complements.
- $\epsilon > 1$ is the elasticity of substitution between cash and deposits, so cash and deposits are substitutes in supplying liquidity.
- η is the elasticity of substitution across banks, generating imperfect competition in deposit markets.

3.2 Bank problem and the equilibrium deposit spread

Each bank chooses its deposit spread s_i to maximize deposit profits $D_i s_i$. The first-order condition is

$$\frac{\partial D_i}{\partial s_i} \frac{s_i}{D_i} = -1. \quad (4)$$

In symmetric equilibrium, the elasticity of demand for bank i 's deposits is

$$\frac{\partial D_i}{\partial s_i} \frac{s_i}{D_i} = \frac{1}{N} \left(\frac{\partial D}{\partial s} \frac{s}{D} \right) - \eta \left(1 - \frac{1}{N} \right). \quad (5)$$

Combining this with the bank's first-order condition gives the key aggregate condition

$$-\frac{\partial D}{\partial s} \frac{s}{D} = 1 - (\eta - 1) \left(1 - \frac{1}{N} \right) \equiv \mathcal{M}. \quad (6)$$

Thus, $\mathcal{M}$ is the model's summary measure of **deposit-market power**. To solve for the spread, the paper derives the elasticity of aggregate deposit demand:

$$-\frac{\partial D}{\partial s} \frac{s}{D} = \left[\frac{1}{1 + \delta^\epsilon \left(\frac{f}{s}\right)^{\epsilon-1}} \right] \epsilon + \left[\frac{\delta^\epsilon \left(\frac{f}{s}\right)^{\epsilon-1}}{1 + \delta^\epsilon \left(\frac{f}{s}\right)^{\epsilon-1}} \right] \rho. \quad (7)$$

In the limiting case $\lambda \rightarrow 0$, the equilibrium deposit spread satisfies

$$s = \delta^{\frac{\epsilon}{\epsilon-1}} \left(\frac{\mathcal{M} - \rho}{\epsilon - \mathcal{M}} \right)^{\frac{1}{\epsilon-1}} f, \quad \text{if } \mathcal{M} > \rho. \quad (8)$$

Interpretation.

- A higher Fed funds rate f makes cash more expensive to hold, so households are less willing to substitute from deposits into cash.
- Deposit demand becomes less elastic, which raises banks' effective market power.
- As a result, banks optimally widen the deposit spread when the Fed funds rate rises.
- The slope $\partial s / \partial f$ – the **deposit spread beta** – is larger when market power $\mathcal{M}$ is higher.

3.3 Financial sophistication and nonswitchers

To allow for financially inattentive households, let α_{ns} be the fraction of nonswitchers. Then the deposit spread still takes the form above, but with $\mathcal{M}$ replaced by

$$\mathcal{M}_{ns} = 1 - (\eta - 1) \left[\frac{1}{\alpha_{ns} + (1 - \alpha_{ns}) \frac{1}{N}} - 1 \right]. \quad (9)$$

Interpretation.

- More nonswitchers raise market power because fewer depositors discipline banks by moving funds elsewhere.
- The deposit spread beta rises both with concentration and with depositor inattentiveness.
- This motivates the empirical use of education, income, and age as additional proxies for market power.

3.4 Lending, wholesale funding, and the deposits channel

The baseline model is extended so banks also have profitable lending opportunities and costly wholesale funding. Bank i solves

$$\Pi_i = \max_{D_i, H_i} \left(f + l_0 - \frac{l_1}{2} L_i \right) L_i - \left(f + \frac{h}{2} H_i \right) H_i - (f - s_i) D_i, \quad (10)$$

where total lending is

$$L_i = H_i + D_i. \quad (11)$$

The optimal wholesale-funding choice is

$$H_i = \frac{l_0}{l_1 + h} - \frac{l_1}{l_1 + h} D_i, \quad (12)$$

so lending is

$$L_i = \frac{l_0}{l_1 + h} + \frac{h}{l_1 + h} D_i. \quad (13)$$

The optimality condition for deposits becomes

$$0 = \frac{\partial \Pi_i}{\partial D_i} = \left(\frac{h}{l_1 + h} \right) (l_0 - l_1 D_i) + s_i \left(1 + \frac{\partial s_i}{\partial D_i} \frac{D_i}{s_i} \right). \quad (14)$$

Interpretation.

- Deposits now matter not only because they generate franchise rents, but also because they fund profitable loans.
- Banks partially offset deposit outflows with wholesale funding, but wholesale funding is costly and therefore imperfectly substitutes for deposits.
- Hence, when the Fed funds rate rises, banks reduce deposits, increase wholesale funding, and on net reduce lending.

3.5 Branch-level betas and within-bank deposit regressions

The paper first estimates branch-specific sensitivities to monetary policy using

$$y_{it} = \alpha_i + \beta_i \text{FF}_t + \varepsilon_{it}, \quad (15)$$

where y_{it} is either the change in the deposit spread or the log change in deposits at branch i .

The main within-bank branch regression is

$$y_{it} = \alpha_i + \zeta_{c(i)} + \lambda_{s(i)t} + \delta_{j(i)t} + \gamma(\text{FF}_t \times \text{Branch-HHI}_i) + \varepsilon_{it}. \quad (16)$$

Interpretation.

- Branch fixed effects absorb permanent branch characteristics.
- County and state-time fixed effects absorb local shocks.
- Bank-time fixed effects absorb all time-varying bank-level lending opportunities.
- Therefore, identification comes from comparing **different branches of the same bank** across counties with different concentration.

3.6 Event study and expected-versus-unexpected rate changes

For weekly deposit-rate data, the event-study specification is

$$y_{it} = \alpha_t + \zeta_{c(i)} + \sum_{\tau=-5}^5 \gamma_\tau \text{Branch-HHI}_{c(i)} \times \text{FF}_{t-\tau} + \varepsilon_{it}. \quad (17)$$

The paper also decomposes Fed funds changes into expected and unexpected components using Fed funds futures and estimates deposit-spread regressions with both parts included.

Interpretation.

- The event study tests whether spread responses occur immediately around FOMC changes.
- The expected-rate specification is powerful because deposits are short-lived, so their prices should react even to fully expected rate changes.
- This helps isolate the effect of the rate itself from informational content in Fed announcements.

3.7 Lending regressions

For bank-county small-business lending, the main specification is

$$y_{jct} = \alpha_{jc} + \delta_{ct} + \beta \text{Bank-HHI}_{j,t-1} + \gamma(\text{FF}_t \times \text{Bank-HHI}_{j,t-1}) + \varepsilon_{jct}. \quad (18)$$

For county-level lending and employment, the paper estimates

$$y_{ct} = \alpha_c + \delta_t + \beta \text{County-HHI}_{c,t-1} + \gamma(\text{FF}_t \times \text{County-HHI}_{c,t-1}) + \varepsilon_{ct}. \quad (19)$$

For bank-level balance-sheet components, the specification is

$$y_{bt} = \alpha_b + \delta_t + \beta \text{Bank-HHI}_{b,t-1} + \gamma(\text{FF}_t \times \text{Bank-HHI}_{b,t-1}) + \varepsilon_{bt}. \quad (20)$$

Interpretation.

- Lending regressions compare **different banks within the same county**, thereby controlling for local lending opportunities.
- If the deposits channel is real, banks funded in more concentrated deposit markets should contract lending more after rate hikes.

3.8 Deposit spread beta as a sufficient statistic

The bank-level spread beta is estimated from

$$y_{it} = \alpha_i + \sum_{\tau=0}^4 \beta_{i\tau} \text{FF}_{t-\tau} + \varepsilon_{it}, \quad (21)$$

where y_{it} is the bank's change in deposit spread and the deposit spread beta is $\sum_{\tau=0}^4 \beta_{i\tau}$.

Interpretation.

- The spread beta summarizes how aggressively a bank can raise deposit spreads when the Fed funds rate rises.
- In the model, this is a sufficient statistic for overall deposit-market power, including concentration and depositor sophistication.
- The paper then links this sufficient statistic to deposit, asset, securities, and loan betas.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the aggregate time-series approach

- Aggregate data clearly show that deposit spreads rise and deposit quantities fall when the Fed funds rate increases.
- But aggregate time-series evidence alone cannot rule out the possibility that monetary tightening reduces banks' **lending opportunities**, which would also reduce deposit-taking.
- Therefore, an aggregate negative correlation between rates and deposits is not by itself a clean causal test of the deposits channel.

4.2 What identifies the deposit-supply effect

- The paper uses **cross-sectional variation in local deposit-market concentration** as observable variation in market power.
- The key prediction is differential: after a Fed funds increase, branches in more concentrated markets should raise deposit spreads more and lose more deposits.
- The core identification design is **within-bank**: compare branches of the same bank across counties with different HHIs.
- This design controls for bank-specific loan demand, balance-sheet conditions, and any other time-varying bank-level shocks through bank-time fixed effects.

4.3 Why the lending results are credible

- For lending, the relevant exposure is **Bank-HHI**, because banks can move funds internally across branches.
- The paper therefore compares different banks **within the same county**, using county-time fixed effects to absorb local loan-demand shocks.
- The fact that local deposit concentration itself does not predict lending responses once bank-level concentration is included supports the assumption that deposits are reallocated inside the bank rather than trapped locally.

4.4 Timing evidence and information shocks

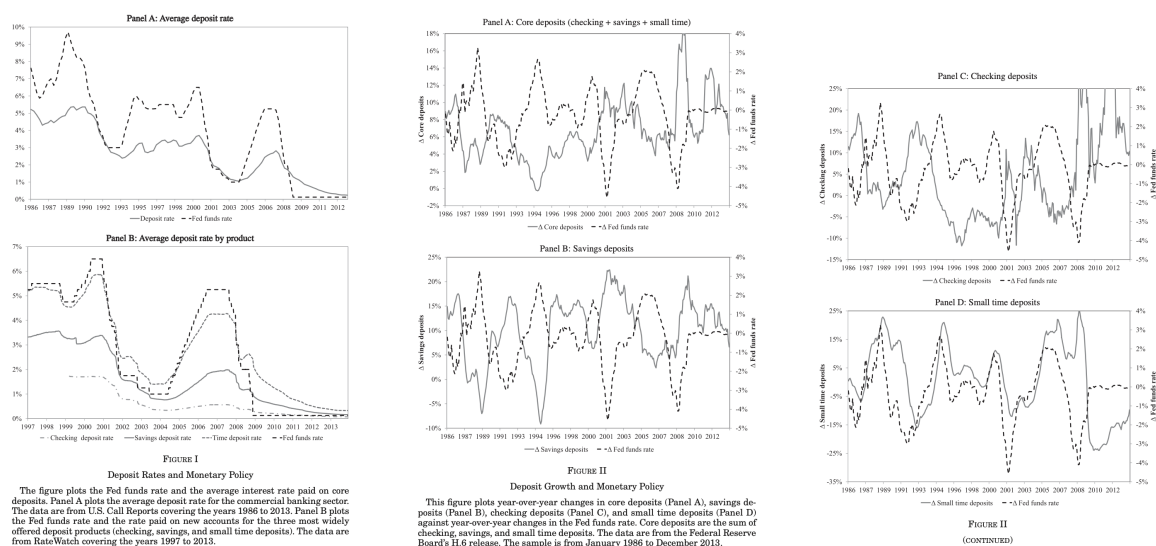
- Weekly event-study results show that the differential response of deposit spreads occurs **immediately** around Fed announcements, with no pre-trend.
- Replacing total Fed funds changes with **expected** and **unexpected** components shows that even expected changes move deposit spreads, which is what the model predicts for short-maturity deposits.
- This is particularly important because it weakens the concern that the results are driven by private information released by the Fed rather than by the change in rates itself.

4.5 Additional credibility and relation to the literature

- Financial sophistication proxies – age, income, and education – work in the same direction as concentration, which is consistent with the model rather than with a narrow market-structure-only story.
- The paper provides an alternative to reserve-based versions of the bank-lending channel and connects monetary policy to the liquidity-premium literature.
- Identification comes mainly from fixed-effects designs, timing, and model-implied heterogeneity, rather than from a traditional excluded instrument.

5 Tables and figures

5.1 Figure 1 and Figure 2: aggregate deposit pricing and quantity facts



Read:

- Figure 1 shows that banks do *not* pass through the Fed funds rate one-for-one into deposit rates. The spread widens when policy tightens.
- The widening is much larger for liquid deposit products such as savings and checking than for small time deposits.
- Figure 2 shows the quantity side: core deposits, especially savings and checking, fall when the Fed funds rate rises.
- The key economic point is that **prices go up while quantities go down**. That is the signature of an inward shift in **deposit supply**, not an outward shift in deposit demand.

5.2 Figure 3 and Table 1: local concentration and sample background

Read:

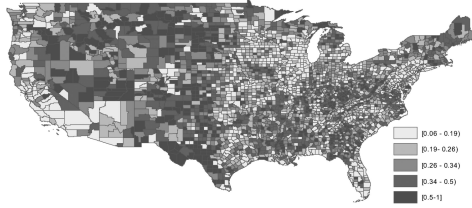


FIGURE III
Concentration in Local Deposit Markets

This map shows the Herfindahl index for each U.S. county. The Herfindahl is calculated each year using the deposit market shares of all banks with branches in a given county and then averaged over the period from 1994 to 2013. The underlying data are from the FDIC.

- Figure 3 shows that county-level deposit-market concentration varies enormously across the United States.
- Table 1 shows that high-HHI counties are smaller and less educated on average, while low-HHI counties are larger and richer.
- This matters because the paper will compare deposit responses across markets with very different local competitive environments.
- The table also shows that the branch, bank, and county samples are large enough to support the fixed-effects identification strategy.

5.3 Figure 4 and Tables 2–3: concentration, deposit spreads, and deposit flows

Read:

- Figure 4 shows nonparametrically that spread betas are higher and flow betas more negative in more concentrated counties.
- Table 2 is the main branch-level spread result: within the same bank, a 100 bps increase in the Fed funds rate raises savings-deposit spreads by about **14 bps more** and time-deposit spreads by about **7 bps more** in high-concentration than in low-concentration markets.
- Table 3 is the main branch-level quantity result: within the same bank, deposits flow out by about **66 bps more** from high-concentration branches after a 100 bps rate increase.

TABLE I
DESCRIPTIVE STATISTICS

	All		Low HHI		High HHI	
	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.
Panel A: County characteristics (2000 census)						
Population	90,845	294,719	150,081	394,457	28,717	85,292
Area (sq. mile)	1,057	2,484	903	1,279	1,217	3,299
Median income (\$)	42,183	9,844	45,816	10,155	38,550	8,021
Over 65 (%)	14.78	4.14	14.18	4.02	15.37	4.17
College degree (%)	16.55	7.81	18.82	8.53	14.27	6.23
Branch-HHI	0.35	0.20	0.20	0.05	0.49	0.20
Obs. (counties)	3,104		1,552		1,552	
Panel B: Branch characteristics (FDIC)						
Deposits (mill. \$)	67.18	878.2	59.47	271.5	74.89	1,212.0
Deposit growth (%)	7.71	25.36	8.58	26.62	6.84	24
Δ FF	-0.25	1.44	-0.26	1.44	-0.24	1.44
Branch-HHI	0.22	0.11	0.15	0.03	0.29	0.11
Obs. (branch $\times$ year)	1,310,111		654,840		655,271	
Panel C: Branch characteristics (Ratewatch)						
Deposits (mill. \$)	142.6	1,108	112.9	408.4	172.3	1,512
Δ Spread (savings)	-0.03	0.49	-0.03	0.49	-0.03	0.49
Δ Spread (time)	0.00	0.37	0.00	0.37	-0.01	0.37
Branch-HHI	0.22	0.09	0.15	0.03	0.29	0.07
Obs. (branch $\times$ quarter)	412,008		205,749		206,259	
Panel D: Bank characteristics (Call Reports)						
Assets (mill. \$)	1,028	21,961	805	12,308	1,230	28,004
Deposits/Liab. (%)	94.19	7.09	93.92	7.21	94.44	6.97
Branches	10.01	81.81	9.05	71.77	10.93	90.32
Bank-HHI	0.24	0.14	0.15	0.04	0.33	0.13
Obs. (bank $\times$ quarter)	558,739		279,364		279,375	
Panel E: Small business lending (NCRC)						
New lending (mill. \$)	5.76	22.99	6.31	24.44	5.22	21.43
New lending (log)	6.88	1.63	6.95	1.65	6.82	1.61
Assets (bill. \$)	132.1	308.2	121.9	306.1	142.4	310.0
Bank-HHI	0.26	0.14	0.18	0.04	0.34	0.15
Obs. (bank $\times$ county $\times$ year)	620,443		310,210		310,233	

Notes. This table provides summary statistics at the county, branch, bank, and county-bank levels. All panels provide a breakdown by high and low Herfindahl (HHI) using the median HHI for the respective sample. Panel A presents county characteristics for all U.S. counties with at least one bank branch. The underlying data are from the 2000 census. Panel B presents data on deposit holdings and deposit growth. The underlying data are from the FDIC from June 1994 to June 2013. Panel C presents data on deposit spreads. The underlying data are from Ratewatch from January 1997 to December 2013. Panel D presents data on bank characteristics. The underlying data are from the Call Reports from 1994 to 2013. Panel E presents small business lending data. The underlying data are from the NCRC for the years 1997 to 2013.

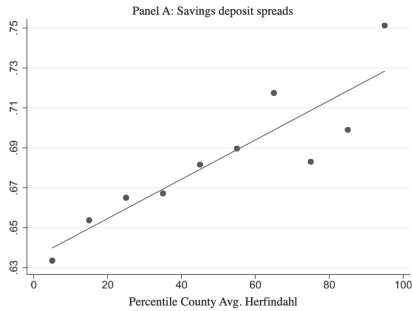


FIGURE IV
Spread and Flow Betas by Market Concentration

This figure shows the relationship between market concentration and the sensitivities of deposit spreads and flows to the Fed funds rate. The figure is constructed in two steps. The first is to estimate spread and flow betas using a time-series regression for each branch i : $\Delta y_{it} = \alpha + \beta_i \Delta FFR_t + \varepsilon_{it}$, where Δy_t is either the change in the deposit spread or the log change in deposits (deposit flow) from date t to $t+1$ and ΔFFR_t is the change in the Fed funds target rate from t to $t+1$. The second step is to average betas by county and then sort counties into 20 bins by market concentration and compute average betas by bin. Panel A shows the results for savings deposit spreads. Panel B shows the results for time deposit spreads. Panel C shows the results for deposit flows. The data for Panels A and B are from Ratewatch covering January 1997 to December 2013. The data for Panel C are from the FDIC covering January 1994 to December 2013.

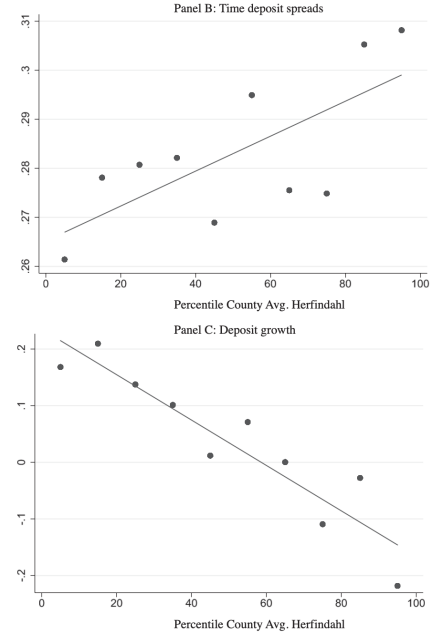


FIGURE IV
(CONTINUED)

TABLE II
DEPOSIT SPREADS AND MONETARY POLICY

	Δ Spread					
	≥ 2 Counties			All		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Savings deposits						
$\Delta FF \times \text{Branch-HHI}$	0.140*** (0.033)	0.101*** (0.031)	0.099*** (0.043)	0.199*** (0.028)	0.155*** (0.026)	0.159*** (0.026)
Bank $\times$ quarter f.e.	Y	Y	N	N	N	N
State $\times$ quarter f.e.	Y	N	N	Y	N	N
Branch f.e.	Y	Y	N	Y	Y	N
County f.e.	Y	Y	Y	Y	Y	Y
Quarter f.e.	Y	Y	Y	Y	Y	Y
Observations	117,683	117,683	117,683	412,008	412,008	412,008
R^2	0.810	0.799	0.559	0.659	0.650	0.645
Panel B: Time deposits						
$\Delta T\text{-bill} \times \text{Branch-HHI}$	0.074*** (0.025)	0.073*** (0.026)	0.156*** (0.037)	0.156*** (0.026)	0.119*** (0.024)	0.119*** (0.023)
Bank $\times$ quarter f.e.	Y	Y	N	N	N	N
State $\times$ quarter f.e.	Y	N	N	Y	N	N
Branch f.e.	Y	Y	N	Y	Y	N
County f.e.	Y	Y	Y	Y	Y	Y
Quarter f.e.	Y	Y	Y	Y	Y	Y
Observations	121,990	121,990	121,990	430,050	430,050	430,050
R^2	0.808	0.796	0.442	0.513	0.492	0.488

Notes: This table estimates the effect of Fed funds rate changes on deposit spreads. The data are at the branch-quarter level and covers January 1997 to December 2013. In columns (1) to (3) the sample consists of banks with branches in two or more counties. In columns (4) to (6) the sample consists of all banks. Δ Spread is the change in branch-level deposit spread, which is equal to the change in the Fed funds target rate minus the change in the deposit rate. Branch-HHI measures market concentration in the county where a branch is located. ΔFF is the change in the Fed funds target rate. $\Delta T\text{-bill}$ is the change in the one-year T-bill rate. Panel A reports the results for savings deposits. Panel B reports the results for time deposits. The data are from Ratewatch. Panel effects (f.e.) are denoted at the bottom of each panel. Standard errors are clustered by county. **, *** indicates significance at the 0.05 and 0.01 levels, respectively.

TABLE III
DEPOSIT GROWTH AND MONETARY POLICY

	Δ log Deposits					
	≥ 2 Counties			All		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta FF \times \text{Branch-HHI}$	-0.661*** (0.254)	-1.008*** (0.331)	-0.826*** (0.246)	-1.827*** (0.198)	-1.796*** (0.242)	-0.963*** (0.212)
Bank $\times$ year f.e.	Y	Y	N	N	N	N
State $\times$ year f.e.	Y	N	N	Y	N	N
Branch f.e.	Y	Y	N	Y	Y	N
County f.e.	Y	Y	Y	Y	Y	Y
Quarter f.e.	Y	Y	Y	Y	Y	Y
Observations	1,150,049	1,150,049	1,150,049	1,310,111	1,310,111	1,310,111
R^2	0.344	0.336	0.025	0.230	0.221	0.025

Notes: This table estimates the effect of Fed funds rate changes on deposit growth. The data are at the branch-year level and cover the years 1994 to 2013. In columns (1) to (3) the sample consists of all banks with branches in two or more counties. In columns (4) to (6) the sample consists of all banks. Deposit growth is the log change in deposits at the branch level. Branch-HHI measures market concentration in the county where a branch is located. ΔFF is the change in the Fed funds target rate. The data are from the FDIC. Panel effects are denoted at the bottom of the table. Standard errors are clustered by county. *** indicates significance at the 0.01 level.

- The economic logic is exactly the model's: more market power means stronger pass-through from the Fed funds rate into deposit spreads and therefore stronger deposit outflows.

5.4 Figure 5 and Table 4: timing and expected-rate evidence

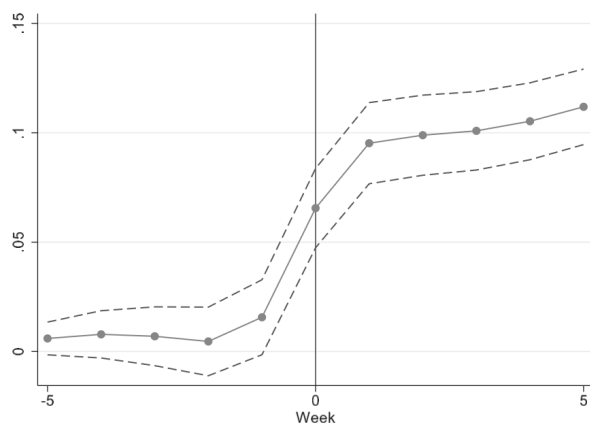


FIGURE V

Deposit Spreads and Monetary Policy (Event Study)

This figure shows the effect of Fed funds rate changes on deposit spreads at a weekly frequency. The figure plots the coefficient sum $\sum_{\tau=-5}^t \gamma_{\tau}$, $t = -5, \dots, 5$ (week 0 corresponds to an FOMC meeting), and associated 95% confidence interval, estimated from the regression $\Delta y_{it} = \alpha_t + \zeta_{c(i)} + \sum_{\tau=-5}^5 \gamma_{\tau} \text{Branch-HHI}_{c(i)} \times \Delta FF_{t-\tau} + \varepsilon_{it}$, where Δy_{it} is the change in the savings deposit spread of branch i from week t to $t+1$, $\Delta FF_{t-\tau}$ is the change in the Fed funds target rate from week $t-\tau$ to $t-\tau+1$, and $\text{Branch-HHI}_{c(i)}$ is the concentration of county $c(i)$ in which branch i is located. The data are from Ratewatch covering January 1997 to December 2013.

TABLE IV
DEPOSIT SPREADS AND EXPECTED CHANGES IN MONETARY POLICY

	ΔSpread					
	≥ 2 Counties			All		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \text{Exp. FF} \times \text{Branch-HHI}$	0.217*** (0.074)	0.148** (0.072)	0.190** (0.076)	0.256*** (0.047)	0.186*** (0.046)	0.190*** (0.045)
$\Delta \text{Unexp. FF} \times \text{Branch-HHI}$	0.114* (0.061)	0.081 (0.056)	0.016 (0.071)	0.200*** (0.040)	0.156*** (0.035)	0.161*** (0.036)
Bank $\times$ quarter f.e.	Y	Y	N	N	N	N
State $\times$ quarter f.e.	Y	N	N	Y	N	N
Branch f.e.	Y	Y	N	Y	Y	N
County f.e.	Y	Y	Y	Y	Y	Y
Quarter f.e.	Y	Y	Y	Y	Y	Y
Observations	117,683	117,683	117,683	412,008	412,008	412,008
R^2	0.810	0.799	0.559	0.659	0.650	0.645

Notes. This table estimates the effect of expected Fed funds rate changes on deposit spreads. The data are at the branch-quarter level from January 1997 to December 2013. In columns (1) to (3) the sample consists of banks with branches in two or more counties. In columns (4) to (6) the sample consists of all banks. The analysis focuses on savings deposits because they have zero maturity. $\Delta \text{Exp. FF}$ is the expected change in the Fed funds rate computed as the Fed Funds target rate minus the three-month Fed funds futures rate at the start of a quarter. $\Delta \text{Unexp. FF}$ is the unexpected change in the Fed funds rate, computed as the difference between the realized change and the expected change. The other variables are defined in Table II. Fixed effects are denoted at the bottom of each panel. Standard errors are clustered by county. *, **, *** indicates significance at the 0.1, 0.05, and 0.01 levels, respectively.

Read:

- Figure 5 shows that the differential increase in deposit spreads at high-HHI branches happens essentially **within a week** of a Fed funds target change.
- There is no meaningful pre-trend, which is strong evidence against slow-moving omitted variables.
- Table 4 shows that **expected** Fed funds changes also increase deposit spreads more in concentrated markets.
- This is crucial because short-maturity deposits should react to the actual rate change whether or not it was anticipated, whereas information-shock stories have a harder time matching this pattern.

5.5 Table 5: financial sophistication as another source of market power

Read:

- Table 5 shows that branches in counties with older populations, lower income, and lower education respond more strongly to Fed funds increases.
- These results line up with the model's extension in which inattentive or unsophisticated depositors make deposit demand less elastic.
- The point is not only that concentration matters, but that the broader concept of **deposit-market power** matters.

	Δ Spread			
	(1)	(2)	(3)	(4)
Panel A: Deposit spreads				
$\Delta FF \times \text{Branch-HHI}$	0.144*** (0.029)	0.086*** (0.029)	0.086*** (0.028)	0.073** (0.029)
$\Delta FF \times \text{Age}$	0.402*** (0.072)		0.026 (0.076)	
$\Delta FF \times \text{Income}$		-0.127*** (0.015)		-0.035* (0.021)
$\Delta FF \times \text{College}$			-0.215*** (0.034)	-0.256*** (0.046)
All f.e.	Y	Y	Y	Y
Observations	411,976	411,976	411,976	411,976
R^2	0.659	0.659	0.659	0.659
Δ log Deposits				
	(1)	(2)	(3)	(4)
Panel B: Deposit growth				
$\Delta FF \times \text{Branch-HHI}$	-1.273*** (0.197)	-1.512*** (0.215)	-1.597*** (0.209)	-1.268*** (0.204)
$\Delta FF \times \text{Age}$	-6.458*** (0.681)			-6.476*** (0.825)
$\Delta FF \times \text{Income}$		0.452*** (0.158)		0.046 (0.244)
$\Delta FF \times \text{College}$			0.936*** (0.310)	-0.114 (0.472)
All f.e.	Y	Y	Y	Y
Observations	1,299,505	1,299,505	1,299,505	1,299,505
R^2	0.231	0.231	0.231	0.231

Notes. This table estimates the effect of financial sophistication on deposit spreads and deposit growth. The data are at the branch-quarter level from January 1997 to December 2013 (Panel A) and the branch-year level from 1994 to 2013 (Panel B). Panel A reports results on savings deposit spreads. Panel B reports results for deposit growth. Age is the county share of the population aged 65 or older. Income is the natural log of county-level median household income. College is the county share of the population with a college degree. All other variables are defined in Table 2. All regressions include state-time, branch, and county fixed effects. Standard errors are clustered by county. *, **, *** indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.

	log(new lending)			
	(1)	(2)	(3)	(4)
$\Delta FF \times \text{Bank-HHI}$	-0.208*** (0.085)	-0.197*** (0.088)	-0.168*** (0.076)	-0.166*** (0.075)
$\Delta FF \times \text{Branch-HHI}$		0.026 (0.016)	0.010 (0.023)	
Time f.e.	Y	Y	Y	Y
County f.e.	Y	Y	Y	Y
Bank f.e.	Y	Y	Y	Y
County-bank f.e.	Y	Y	N	N
County-time f.e.	Y	N	N	N
Observations	620,443	620,443	620,443	620,443
R^2	0.830	0.815	0.246	0.246

Notes. This table estimates the effect of the deposits channel on new small business lending. The data are at the bank-county-year level from 1997 to 2013. Log(new lending) is the log of the total amount of new small business loans originated by a given bank in a given county and year. Bank-HHI is the bank-level average of Branch-HHI using lagged deposit shares across branches as weights. All other variables are defined in Table 1. The regression includes a control for Bank-HHI (coefficient not shown). The data are from the NCB. Fixed effects are denoted at the bottom. Standard errors are clustered by bank and county. ** indicates significance at the 0.05 level.

	log(new lending)		Δ Employment		Δ Wage bill	
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta FF \times \text{County-HHI}$	-0.093*** (0.025)	-0.097*** (0.028)	-0.014*** (0.003)	-0.008*** (0.003)	-0.011*** (0.003)	-0.009*** (0.004)
$\Delta FF \times \text{Branch-HHI}$		0.003 (0.009)		-0.004*** (0.001)		-0.001 (0.001)
Time f.e.	Y	Y	Y	Y	Y	Y
County f.e.	Y	Y	Y	Y	Y	Y
Observations	54,107	54,107	54,107	54,107	54,107	54,107
R^2	0.948	0.948	0.201	0.202	0.272	0.272

Notes. This table estimates the effect of the deposits channel on new small business lending and employment. The data are at the county-year level covering the years 1997 to 2013. Log(new lending) is the log of the total amount of new small business loans originated by county per year. Δ Employment is the change in total employment by county and year. Δ Wage bill is the change in the total wage bill by county and year. County-HHI is county-level average of Bank-HHI using one-year lagged lending shares across banks as weights. All other variables are defined in Table 1. The data are from the NCB and the Bureau of Labor Statistics. Fixed effects are denoted at the bottom. Standard errors are clustered by county. *, **, *** indicates significance at the 0.05 and 0.01 levels, respectively.

5.6 Table 6 and Table 7: lending and real activity

Read:

- Table 6 shows that after a 100 bps Fed funds increase, a one-standard-deviation increase in bank-level concentration reduces new small-business lending by roughly **291 bps**.
- Importantly, local branch concentration does not explain lending once bank-level funding concentration is included, which supports the internal-capital-market assumption.
- Table 7 aggregates to the county level and shows that counties served by more exposed banks experience lower **new lending**, **employment growth**, and **wage-bill growth**.
- So the deposits channel is not just about deposit pricing; it propagates into local credit supply and real activity.

5.7 Table 8: bank balance-sheet adjustment

Read:

- Table 8 shows how banks absorb deposit outflows on the liability and asset sides of the balance sheet.
- Banks in more concentrated deposit markets lose more deposits and raise deposit spreads more when rates rise.
- They partly replace lost deposits with **wholesale funding**, but the replacement is incomplete.
- On the asset side, they reduce **total assets**, **securities**, and **loans**, including both real-estate and C&I lending.
- This is exactly the model's mechanism: deposits are special funding, so a contraction in deposits translates into a contraction in lending.

TABLE VIII
DEPOSITS CHANNEL, BANK LIABILITIES, AND LENDING (BANK-LEVEL RESULTS)

	Δ Total deposits (1)	Δ Deposit spread (2)	Δ Savings deposits (3)	Δ Time deposits (4)	Δ Wholesale funding (5)	Δ Total liabilities (6)
Panel A: Liabilities						
Δ FF $\times$ Bank HHI	-1.478*** (0.144)	0.076*** (0.010)	-1.209*** (0.242)	-2.144*** (0.212)	2.438*** (0.945)	-1.280*** (0.138)
Bank f.e.	Y	Y	Y	Y	Y	Y
Quarter f.e.	Y	Y	Y	Y	Y	Y
Observations	558,502	558,502	558,502	558,502	558,502	558,502
R^2	0.162	0.400	0.078	0.170	0.033	0.175
	Δ Total assets (1)	Δ Cash (2)	Δ Securities (3)	Δ Total loans (4)	Δ Real estate loans (5)	Δ C&I loans (6)
Panel B: Assets						
Δ FF $\times$ Bank HHI	-1.204*** (0.123)	-2.477*** (0.666)	-0.952*** (0.335)	-0.462*** (0.150)	-0.824*** (0.196)	-0.913*** (0.350)
Bank f.e.	Y	Y	Y	Y	Y	Y
Quarter f.e.	Y	Y	Y	Y	Y	Y
Observations	558,502	558,502	558,502	558,502	558,502	558,502
R^2	0.176	0.051	0.063	0.223	0.176	0.059

Notes. This table estimates the effect of the deposits channel on bank-level outcomes. The data are at the bank-quarter level and cover all commercial banks from January 1994 to December 2013. Panel A examines the change in bank liabilities. Δ Total deposits, Δ Savings deposits, Δ Time deposits, Δ Wholesale funding, and Δ Total liabilities are calculated as quarterly log changes. Δ Deposit spread is the change in the Fed funds target rate minus the change in the annualized deposit rate (computed as total domestic deposit expense divided by total domestic deposits) over a quarter. Panel B examines the change in bank assets. Δ Total assets, Δ Cash, Δ Securities, Δ Total loans, Δ Real estate loans, and Δ C&I loans are calculated as quarterly log changes. All other variables are defined in Tables II and VI. Fixed effects are denoted at the bottom. Standard errors are clustered by bank. *** indicates significance at the 0.01 level.

5.8 Figure 6, Figure 7, Table 9, and Figure 8: spread beta, aggregate effects, and liquidity premium

Read:

- Figure 6 shows that the bank-level **deposit spread beta** is a powerful sufficient statistic for exposure to monetary policy: banks with larger spread betas have larger deposit, asset, securities, and loan contractions.
- Table 9 quantifies these relationships. For all banks, the spread beta strongly predicts more negative deposit, asset, securities, and loan betas; the relationships are even stronger for large banks.
- Figure 7 focuses on the largest 5% of banks and shows that the mechanism is strong even for the institutions that matter most for the aggregate time series.
- Using large-bank estimates, the paper computes that a typical 400 bps hiking cycle reduces deposits by about **1,458 bps** and lending by about **995 bps**, or **\$1.35 trillion** and **\$763 billion**.
- Figure 8 shows that the aggregate deposit spread moves almost one-for-one with the T-bill liquidity premium. The deposits channel therefore helps explain not only bank lending but also the **supply of liquid assets** in financial markets.

6 Short summary

- **Method.** Build a model of deposit-market power and test it with within-bank, within-county, and timing-based empirical designs.
- **Identification.** Use local deposit-market concentration and bank fixed effects to isolate a direct effect of monetary policy on deposit supply, then trace this effect into lending.

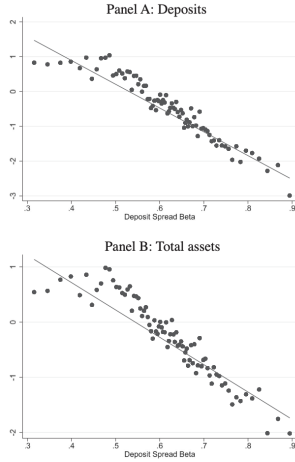


FIGURE VI
Spread Beta, Deposits, and Lending (All Banks)

This figure shows the relationship between exposure to the deposits channel and bank-level outcomes. The figure is constructed in two steps. The first is to estimate bank-level exposure to the deposits channel as the sensitivity of a bank's deposit rate to the Fed funds rate ("spread beta"). The second step is to compute the corresponding sensitivity for bank-level outcomes ("flow beta"). The third step is to sort banks by spread beta into one hundred bins and compute the average spread and flow beta by bin. Panel A shows the results for total deposits, Panel B shows the results for total assets, Panel C shows the results for securities, and Panel D shows the results for total loans. The sample is all U.S. commercial banks from 1994 to 2013 (11,091 banks).

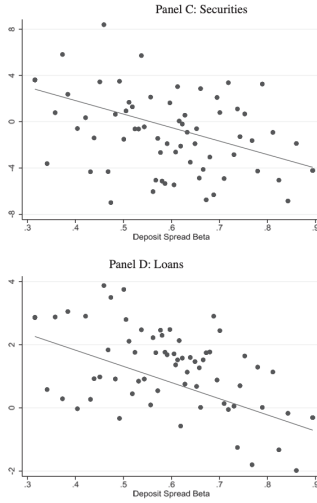


FIGURE VII
(CONTINUED)

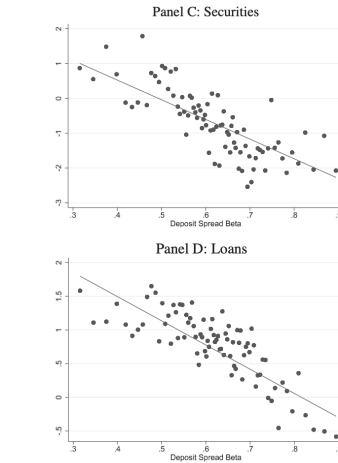


FIGURE VII
(CONTINUED)

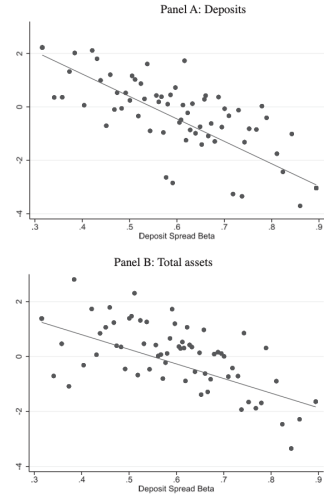


FIGURE VII
Spread Beta, Deposits, and Lending (Large Banks)

This figure shows the relationship between exposure to monetary policy and bank outcomes for large banks. Large banks are banks at or above the 95th percentile of the bank size distribution (555 banks). The figure is constructed the same way as Figure VI. Panel A shows the results for total deposits, Panel B shows the results for total assets, Panel C shows the results for securities, and Panel D shows the results for total loans.

	Deposit- β (1)	Assets- β (2)	Securities- β (3)	Loans- β (4)	RE loans- β (5)	C&I loans- β (6)
Panel A: All banks						
Spread- β	-3.610*** (0.203)	-2.555*** (0.187)	-3.569*** (0.511)	-2.141*** (0.204)	-2.493*** (0.254)	-3.187*** (0.517)
Observations	11,091	11,091	11,091	11,091	11,091	11,091
R^2	0.132	0.085	0.023	0.045	0.042	0.018
Panel B: Large banks						
Spread- β	-5.978*** (0.615)	-4.027*** (0.546)	-8.149*** (1.806)	-4.694*** (0.796)	-5.304*** (0.814)	-6.278*** (1.431)
Observations	555	555	555	555	555	555
R^2	0.318	0.209	0.111	0.171	0.196	0.104

Notes: This table analyzes the relationship between bank market power and bank-level outcomes. The analysis covers U.S. commercial banks from 1994 to 2013. Panel A covers all banks and Panel B covers the largest 5% of banks by assets. We measure bank market power using Spread- β , which is estimated as the sensitivity of a bank's deposit rate to changes in the Fed funds rate. We compute the corresponding Flow- β as the sensitivity of bank-level outcomes to changes in the Fed funds rate. We estimate Flow- β for deposit growth (column (1)), asset growth (column (2)), log change in security holdings (column (3)), loan growth (column (4)), real estate (RE) loan growth (column (5)) and commercial and industrial (C&I) loan growth (column (6)). We report coefficients from regressing Flow- β on Spread- β . We report robust standard errors. ***indicates significance at the 0.01 level.

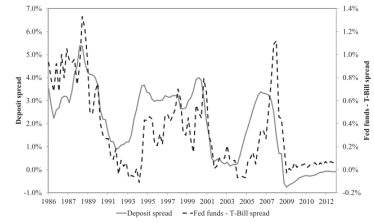


FIGURE VIII
The Aggregate Deposit Spread and the Liquidity Premium

This figure plots the aggregate deposit spread against the T-bill liquidity premium. The deposit spread is equal to the Fed funds rate minus the value-weighted average deposit rate paid by banks, computed from the quarterly Call Reports. The T-bill liquidity premium is equal to the Fed funds rate minus the three-month T-bill rate. Both the Fed funds rate and T-bill rate are calculated as quarterly averages. The data are from January 1986 to December 2013.

- **Application.** U.S. bank branches, banks, counties, and aggregate data from the mid-1980s to 2013/2014.
- **Main result.** When the Fed funds rate rises, banks with more deposit-market power widen spreads more, lose more deposits, and contract lending more. This deposits channel can plausibly account for the full balance-sheet transmission traditionally attributed to the bank-lending channel.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes a new monetary-policy transmission mechanism: the **deposits channel**.
- The key idea is that the Fed funds rate changes banks' effective market power over deposits.
- This mechanism has large effects on deposit spreads, deposit quantities, bank lending, and the supply of liquid assets.

7.2 Economic intuition

- When rates are low, banks compete strongly with cash as an alternative source of liquidity, so they cannot charge large deposit spreads.
- When rates rise, cash becomes relatively expensive, households' demand for deposits becomes less elastic, and banks can widen spreads.
- Since deposits are a uniquely stable source of bank funding, shrinking deposits also shrinks lending.
- This same contraction in deposits also reduces the private supply of liquid assets, helping explain movements in the liquidity premium.

7.3 Takeaway

This paper is important because it moves the analysis of monetary policy away from the traditional reserve-based story and toward **deposit franchise power**. The broad lesson is that bank market power in deposit markets is not a side issue: it is a quantitatively important part of how monetary policy affects credit supply, bank balance sheets, and financial-market liquidity.